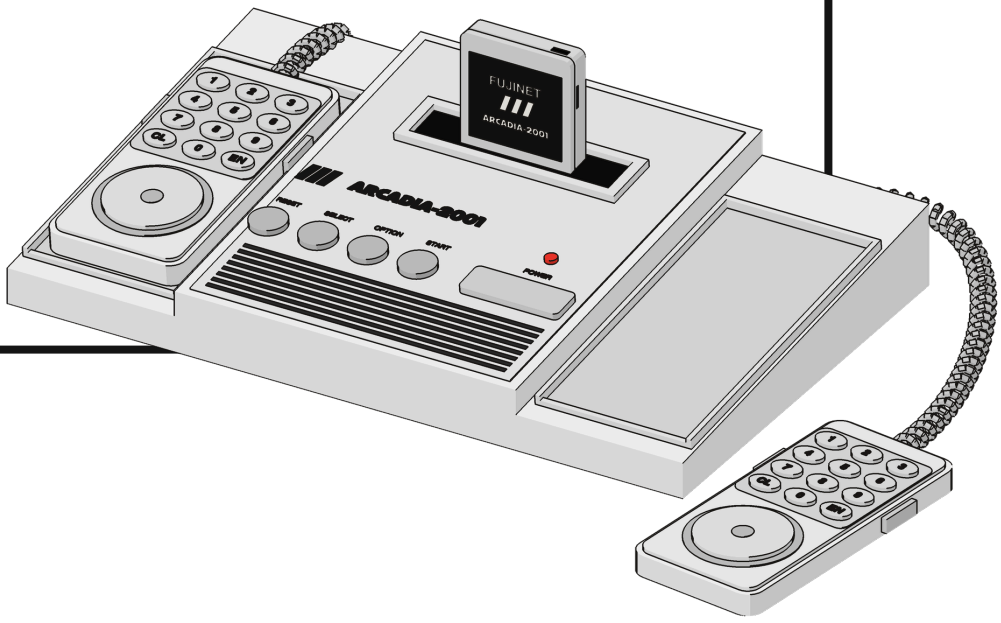
ARCADIA-2001

WITH FUJI NET

NETWORK ADAPTER CARTRIDGE



OWNER'S MANUAL

MODEL NO. FN2001-OM

IN A HURRY?

1. With the console's POWER OFF, push the FujiNet cartridge into the slot, label toward you. Plug a USB-C phone charger into its top.
2. Turn the television to channel 3 or 4 and the switch box to GAME. Press POWER ON.
3. SELECT NETWORK: move to your WiFi network with the left hand controller's disc and press the Action button.
4. Spell the password on the character grid – disc to move, Action button to type, ENTER when done.
5. On FujiNet hosts, open APPS.IRATA.ONLINE, then ARCADIA, and choose a game.
6. To come back: RESET on the back of the cartridge, then the console's RESET.

FujiNet Owner's Manual for the EMERSON ARCADIA-2001. First edition, 2026.

Typeset after the 1982 Emerson *ARCADIA-2001 Owner's Manual* and the Emerson game instruction booklets of the same year, as a tribute. FujiNet is an independent project and is not affiliated with, endorsed by, or sponsored by Emerson Radio Corp. "Emerson" and "Arcadia 2001" are trademarks of their owners and are used here to identify the console this manual is about. The games named in it are the FujiNet project's own, except M.U.L.E., a trademark of its owner, and the games of Section 17, which belong to their makers.

This manual describes the FujiNet cartridge for the ARCADIA-2001 as it is meant to work. The cartridge is new, and the drawings of it and of the console are the manual's own; the screens are the cartridge's own programs, captured running in the FujiNet MAME. Where today's firmware and games still differ, the differences are listed in the manual's errata, kept with its sources.

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1. UNPACKING YOUR FUJINET CARTRIDGE

INTRODUCTION

Welcome to the wonderful world of the FujiNet cartridge for the EMERSON ARCADIA-2001. Your micro-computer was made in 1982 to play the cartridges at your Emerson dealer. With FujiNet it can also play games kept on the Internet, on a memory card, or on a computer in your home – and play some of them against people far away. We suggest that you read this Owner's Manual before setting it up.

WHAT IS INCLUDED

When you take the FujiNet cartridge from its box, you will have:

– The FujiNet cartridge
– This Owner's Manual



FIG. 1
THE FUJINET CARTRIDGE

The FujiNet cartridge goes in the cartridge slot like any other game. Inside it is a complete network adapter: a WiFi radio, a memory-card reader and two small computers of its own.

WHAT YOU ALSO NEED

– **An EMERSON ARCADIA-2001** (model ADV-1), with its two hand controllers. FujiNet is made for the **NTSC** (North American) console and television sets. The Arcadia was also sold under many other names in other countries; FujiNet has not yet been tried in those.
– **Your WiFi network's name and password.** FujiNet uses the 2.4 GHz band, which nearly every home router offers.

You may also want, though you do not need them to start:

– **A microSD memory card**, formatted FAT32 (most cards come that way), to keep games on.
– **A USB-C phone charger and cable**, for extra power. We recommend one; see Section 4.

NOTE: FujiNet makes no changes to your console. Take the cartridge out and your ARCADIA-2001 is exactly as it was.

2. KNOW YOUR CARTRIDGE

The cartridge has two faces. The **label** faces you, toward the front of the console, just as a game's label does. Everything you can press or see light up is on the **back**, the face toward your television. The top edge, which stands up out of the console, has the socket for power, and the right-hand edge the slot for a memory card.

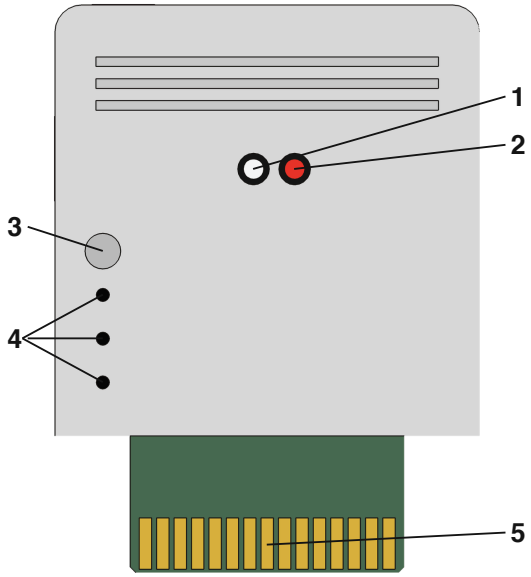


FIG. 2
THE BACK OF THE CARTRIDGE, TOWARD THE TELEVISION

1) STATUS LIGHT
White when FujiNet is connected to your WiFi network; flickers orange while it is busy reading the network or the memory card, or sending a game to the console; dark until it has joined a network.

2) GAME LIGHT
Red whenever the cartridge has power and its game chip is running.

3) RESET BUTTON
Restarts the cartridge itself: both of its small computers. It is the way back to the FujiNet CONFIG from a game (Section 5).

4) SERVICE HOLES
Three small holes, for loading new firmware into the cartridge. Do not push anything into them in everyday use.

5) CONTACTS
The cartridge's connector. Keep it clean, and do not touch the gold contacts.

THE TOP AND THE SIDE

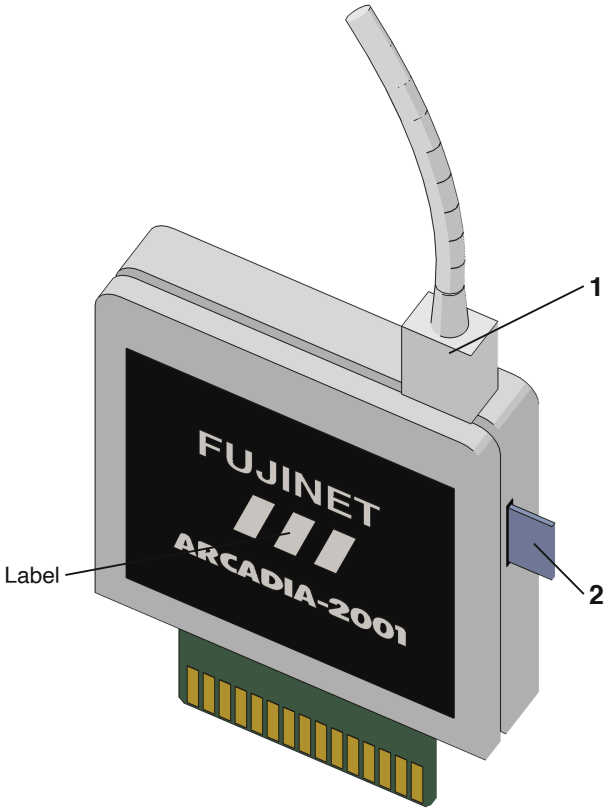


FIG. 3
THE TOP EDGE AND THE RIGHT-HAND SIDE, WITH A CHARGER AND A MEMORY CARD PLUGGED IN

1) USB-C SOCKET
The cartridge can run on power from the console, and a USB-C phone charger plugged in here powers it too (Section 4). Power from the charger never flows back into the console, so it is safe to leave it plugged in. The same socket is used to load new firmware.

2) MEMORY CARD SLOT
A microSD card pushes into the slot in the right-hand edge, contacts first, its printed side toward the label. Push it in until it clicks; push again to let it out.

A LOOK INSIDE

You never need to open the cartridge, but if you are curious, this is the board inside it, seen from the back.

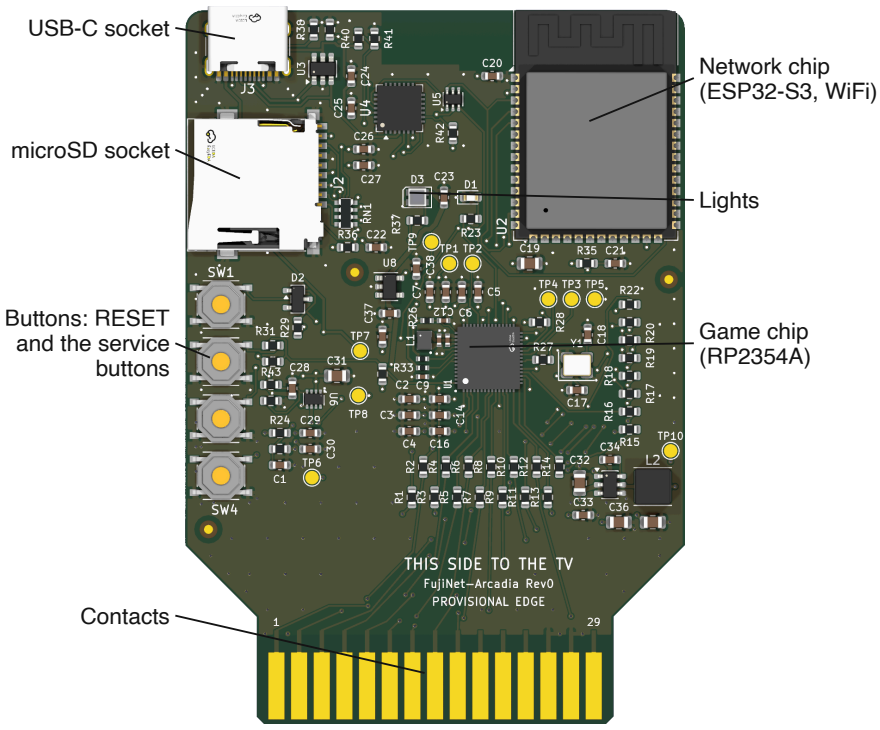


FIG. 4
THE FUJINET ARCADIA BOARD, REV0, FROM THE BACK. THE NETWORK CHIP DOES THE WIFI, THE MEMORY CARD AND THE TALKING TO SERVERS; THE GAME CHIP ANSWERS THE CONSOLE AND PLAYS THE PART OF EACH GAME'S CARTRIDGE.

3. HOW IT WORKS

Your ARCADIA-2001 has no network port, no keyboard and no disk drive, and its cartridge slot has no wire that says “write”. It carries the address the console wants to read, the eight data lines that bring the answer back, and power – nothing else. The console can **read** a cartridge, but it has no way to send it anything at all. A FujiNet cartridge puts it on the Internet anyway. Here is the trick.



FIG. 5
THE 8K THE CONSOLE SEES IN THE CARTRIDGE SLOT, WHILE CONFIG OR A FUJINET GAME IS RUNNING

The console sees a cartridge as two blocks of 4,096 bytes it can look at. FujiNet’s game chip watches **every** look: which address the console reads, at the moment it reads it. So FujiNet sets aside the last pages of the second block. Reading in the **questions** pages is how the console speaks – the address it reads spells the byte it means. The **answers** pages are where FujiNet puts what it found out, and the console reads them like any other part of a cartridge.

	THE CONSOLE, 1982	THE CARTRIDGE, 2026
Computers	one Signetics 2650, about 900,000 steps a second	two (the game chip and the network chip), each running over a hundred times faster
Memory	768 bytes, most of them the picture	about a million bytes, and 8 million more beside
Storage	the 8K a cartridge shows it	18 million bytes of its own, and the memory card
Network	none	WiFi, and through it the whole Internet

HOW A QUESTION GETS ANSWERED

1 **ASK**
The program reads its question out of the questions pages, one byte at a time: to say 65, it reads the 65th place in the page. Nothing crosses the connector but the address, and the game chip catches each one as it goes by.

2 **PASS IT ON**
The game chip hands the question to the network chip beside it.

3 **FETCH**
The network chip does the work a 1982 console never could: it joins your WiFi, reaches the server, or reads the memory card.

4 **ANSWER**
The answer goes into the answers pages. To the console it looks as though it had been printed in the cartridge all along. It simply reads it.

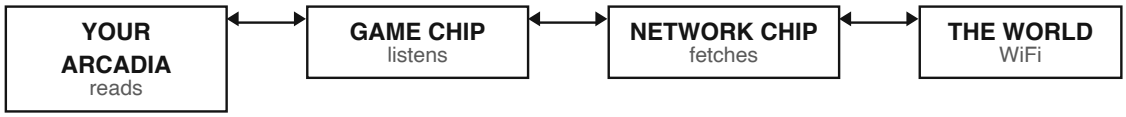


FIG. 6
THE WAY A QUESTION GOES, AND THE ANSWER COMES BACK

WHOLE GAMES

When you load a game, FujiNet sends all of it – up to 8,192 bytes, as large as any ARCADIA-2001 cartridge – into the game chip's own memory. Then CONFIG steps aside and the cartridge becomes that game's cartridge. The console starts the game just as it would at power-on, and never knows the difference.

Programs written for FujiNet may be larger. M.U.L.E. is 51,200 bytes: the cartridge shows it to the console a 2K **page** at a time, in a window in the first block, and turns to another page whenever the program asks. A FujiNet program can be up to 71,680 bytes in all.

4. INSERTING THE CARTRIDGE

Connect your ARCADIA-2001 to the television as its own Owner's Manual shows: the output cable into the antenna switch box, the switch box at **GAME**, and the television on channel 3 or 4, whichever the switch on the back of the console is set to.

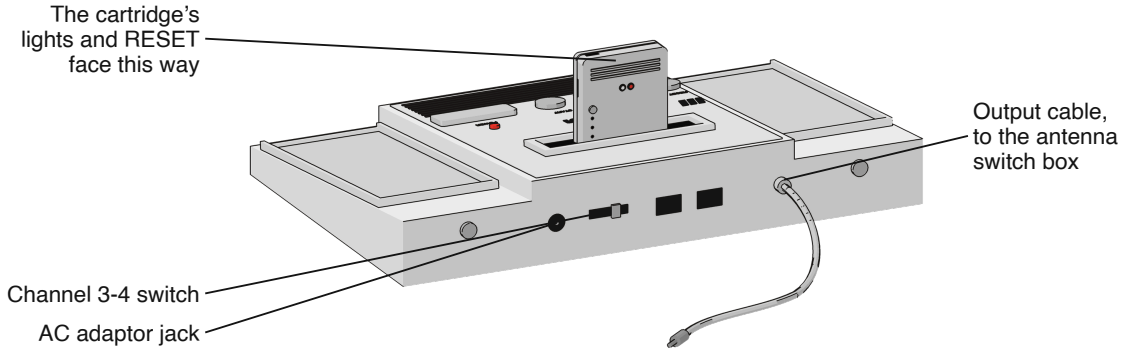


FIG. 7
THE ARCADIA-2001 FROM BEHIND

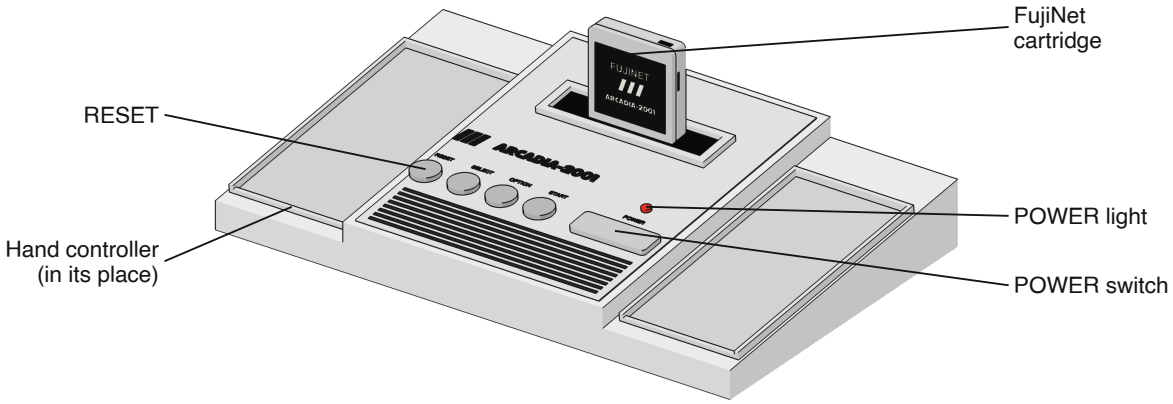


FIG. 8
THE CARTRIDGE IN THE ARCADIA-2001

OPERATION

1. Be sure the **POWER** switch is in OFF position (red POWER light OFF).
2. Hold the FujiNet cartridge so the label faces you and reads right side up, the contacts down.
3. Insert the cartridge into the cartridge slot, firmly but do not force it. The top edge, with its USB-C socket, stays standing out of the slot.
4. Turn on the television and set its volume to a comfortable level.
5. Press the **POWER** switch to ON. The red light glows, and the FujiNet CONFIG appears.

CAUTION: Be sure the POWER light is off before the cartridge is inserted or removed. To remove it, just pull straight up.

The FujiNet CONFIG is worked with the **left** hand controller. Its first screen comes up in a moment; the first time, it asks for your WiFi network (Section 6).

EXTRA POWER

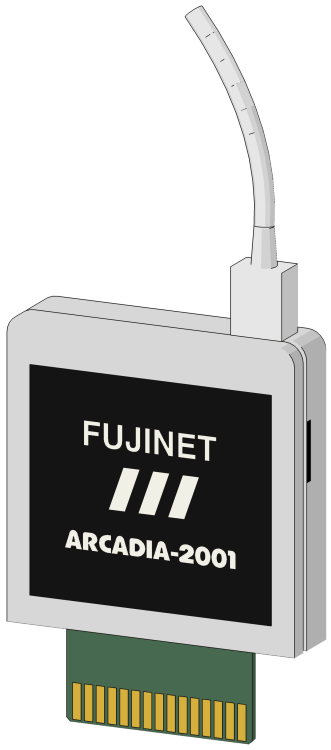


FIG. 9
A USB-C CHARGER PLUGGED IN

The cartridge can run on power from the console, but its WiFi radio needs a good deal more, in short bursts, than the games it replaces: a quarter to two-thirds again of what the whole console draws. The ARCADIA-2001's AC adaptor was never meant to supply that.

A **USB-C phone charger** plugged into the top of the cartridge supplies it instead. The cartridge takes what it needs from the charger first and never sends any of it back to the console. We recommend one whenever you play.

IMPORTANT: Plug the charger into the cartridge **before** you turn the console on, so the cartridge is already awake when the console first looks at it.

With a charger plugged in, FujiNet stays awake and connected while the console is off, so it is ready again the moment you turn the console back on – with the game it last loaded still in it (Section 5 tells how to get back to CONFIG).

5. USING THE HAND CONTROLLER

This micro-computer has two hand controllers, fixed to it by coil cords. The FujiNet CONFIG listens to the **left** one. Lift it out of its place, the keyboard toward the top.

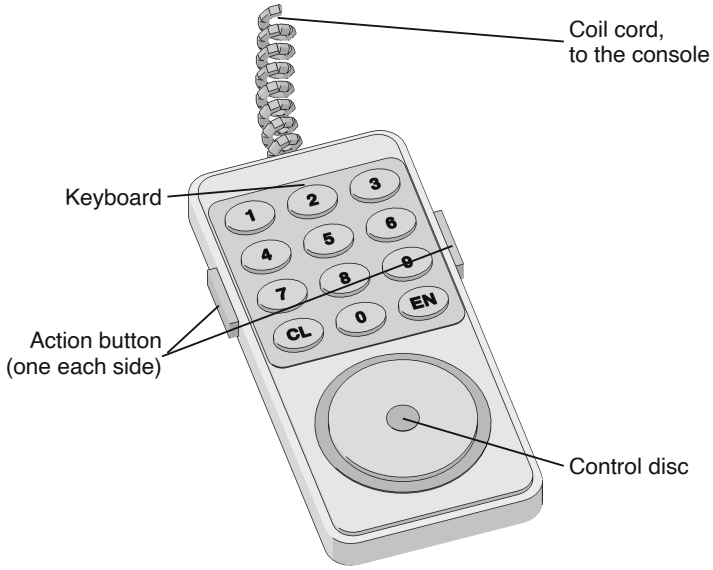


FIG. 10
THE HAND CONTROLLER

Every hand controller has a **keyboard** with 12 buttons, an **Action button** on each side (they do the same thing, for the convenience of either a right- or a left-handed player), and a **control disc**. Press the edge of the disc toward where you want to go – or, with a joy stick screwed into it, push the stick that way. To stop, just let go.

IMPORTANT: On the ARCADIA-2001 the Action buttons and the [2] button are one and the same: the console cannot tell them apart. So in CONFIG and in every FujiNet game, [2] does what the Action button does.

IN THE FUJINET CONFIG

A bar across the screen marks the line you are on. The bottom two lines of every CONFIG screen say which buttons do what there, as **7 WIFI** or **FIRE PICK** – **FIRE** is the Action button.

EVERYWHERE IN CONFIG

▲▼ or [8]	Move down and up the list. [8] goes down only, and from the bottom round to the top. Hold to repeat.
[ACTION], [2] or [ENTER]	Choose the line the bar is on: open it, boot it, pick it.
[CLEAR]	Go back.

The other buttons do the jobs each screen's bottom lines name: on the list of hosts, [1] renames a host, [3] shows FujiNet's INFO, [7] chooses a WiFi network and [9] opens the Game Lobby; in the browser, ◀▶ or [4] and [6] turn the page, [1] goes up a folder, [5] copies and [7] sets a filter. Sections

6 to 11 show each screen. CONFIG does not use the console's START, SELECT and OPTION buttons; the games you load decide what every button does.

KEYBOARD OVERLAY

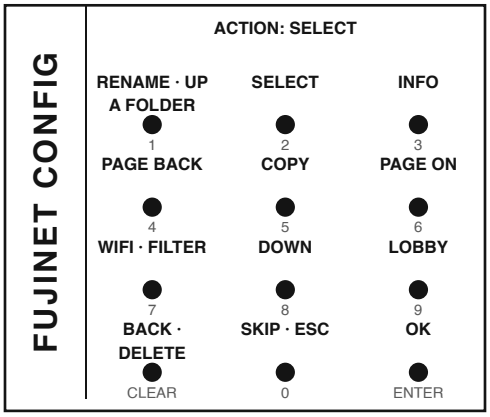


FIG. 11
OVERLAY

ARCADIA-2001 games come with mylar overlays, slipped over the keyboard to show what each button does. This is the FujiNet CONFIG's. Copy it onto a blank overlay, or keep this page beside you while you learn.

Each dot is a button, with its job lettered over it; the button's own name is under it in grey. Where a button has two jobs, the first is on FujiNet hosts and the second in the browser or on the character grid.

GETTING BACK TO CONFIG

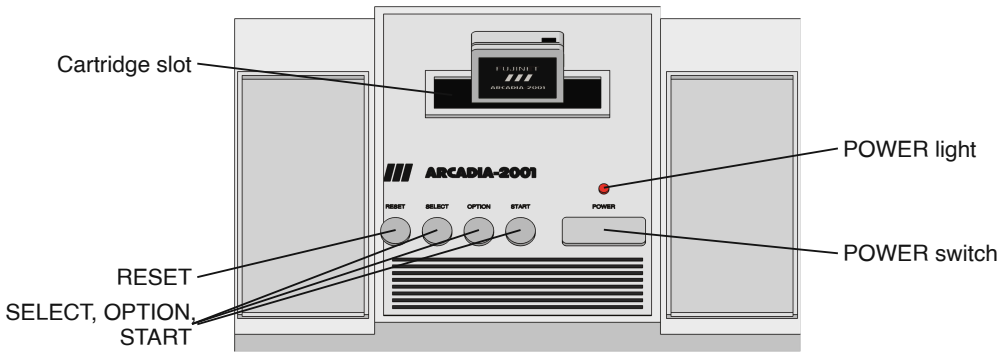


FIG. 12
THE CONSOLE'S BUTTONS

The ARCADIA-2001 has no wire to tell a cartridge to start over. The console's **RESET** button restarts the game you are playing, not CONFIG. To leave a game and go back to CONFIG:

1. Press the **RESET** button on the back of the FujiNet cartridge (Section 2).
2. Press the console's **RESET** button.

CONFIG comes back, with your hosts and WiFi just as you left them.

Turning the console's POWER OFF and ON again does the same, when the cartridge runs on the console's power. With a USB-C charger plugged into the cartridge, the cartridge stays awake while the console is off and keeps the game it last loaded; use the RESET button on the cartridge.

6. JOIN YOUR NETWORK

Each time you turn the console on, CONFIG says **FUJINET CONFIG** and **WAITING FOR LINK** while the cartridge's network chip wakes up – a few seconds, never more than fifteen. Press any button to hurry it along.

If FujiNet already knows your network, it joins it by itself and goes straight to the list of hosts (Section 7). The first time, it has no network to join yet, so it shows **SELECT NETWORK**. **SCANNING** stays on the screen while it listens for WiFi networks nearby; then it lists what it heard.

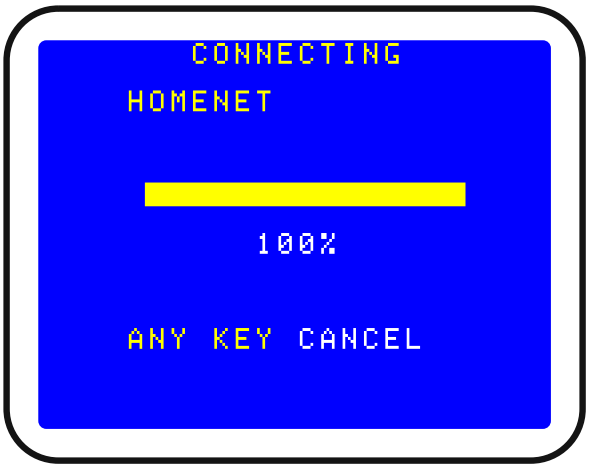


SELECT NETWORK

FIG. 13

Up to seven networks are listed, the strongest first, with a block at the right of each: the taller the block, the stronger FujiNet hears that network. If yours is not in the list – some routers keep their names hidden – choose **OTHER NETWORK** at the bottom; FujiNet asks for the network's name first, and then for its password. [CLEAR] listens again.

1. Move the bar to your network's name.
2. Press the Action button. The character grid appears, titled **PASSWORD**.
3. Spell your network's password (see *The Character Grid*, next).
4. Press [ENTER]. FujiNet says **SAVING**, then **CONNECTING**.



CONNECTING

FIG. 14

CONNECTING shows the network's name and a bar that fills as FujiNet tries to join, for up to about forty seconds. When it has joined, three rising notes sound, the status light on the cartridge turns white, and the list of hosts appears. If the password was not right, or the router is too far away, it says **CONNECT FAILED** and goes back to SELECT NETWORK to try again. Any button while it is trying goes back at once.

NOTE: You can skip WiFi altogether: press [0] at SELECT NETWORK. FujiNet goes to the list of hosts and still plays games from its memory card.

THE CHARACTER GRID

CONFIG shows a grid of characters whenever it needs words: a password, a network's name, a host's name, or a filter. The top line says what it wants; the line under it is what you have typed so far, and beside that, how many characters you have typed of how many it can take.

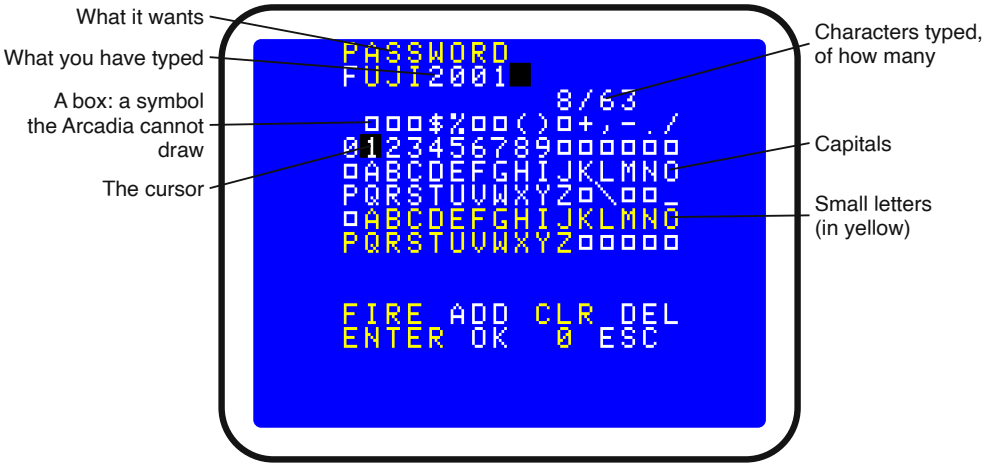


FIG. 15
THE CHARACTER GRID, WITH A PASSWORD TYPED

ON THE CHARACTER GRID

◀▲▼▶, or 4 6 8	Move the cursor from character to character. It goes round at the edges.
ACTION or 2	Type the character the cursor is on.
CLEAR	Erase the last character.
ENTER	Accept what you typed.
0	Leave without changing anything.

The ARCADIA-2001 has no small letters and few symbols. The grid shows small letters as capitals in **yellow**, and every symbol the Arcadia cannot draw as a box; move the cursor onto a box and it shows which symbol it is. What you type is still exactly what you chose.

IMPORTANT: Passwords care about capitals and small letters. Watch the colour: white capitals are capitals; yellow ones are small letters.

A password can be up to 63 characters, a network's name up to 32, a host's up to 31 and a filter up to 15. Only the last 15 show on the line; the count beside them says how many there are.

7. HOSTS AND THE BROWSER

A **host** is a place where games are kept: the memory card in the cartridge, a server on the Internet, or a computer in your home. FujiNet remembers eight of them, and lists them on the **FUJINET HOSTS** screen every time you turn on.

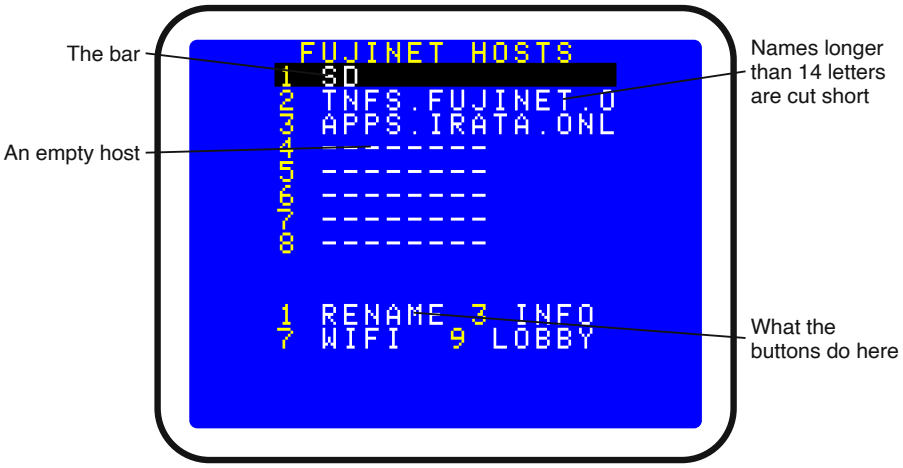


FIG. 16
FUJINET HOSTS, AS IT COMES

A new FujiNet comes with three hosts already filled in:

– **SD** – the microSD card in the side of the cartridge.
– **TNFS.FUJINET.ONLINE** – the FujiNet project's own library.
– **APPS.IRATA.ONLINE** – the library that keeps the ARCADIA-2001's games.

The other five show a row of dashes. Move the bar to a host and press the Action button (or ENTER) to look inside it.

ON FUJINET HOSTS

ACTION or ENTER	Open the host the bar is on.
1	Rename: fill in, or change, the host the bar is on.
3	FujiNet's INFO (Section 10).
7	Choose a WiFi network again (Section 6).
9	The Game Lobby (Section 11).

ADDING A HOST



FIG. 17
NAMING A HOST

1. Move the bar to an empty host – a row of dashes.
2. Press 1. The character grid appears, titled **HOST NAME**.
3. Spell the server's name or address, for example **192.168.1.20**.
4. Press ENTER.

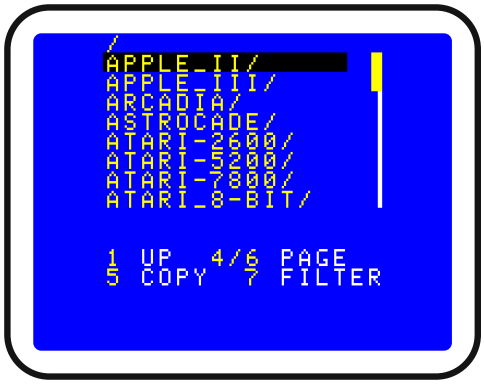
To change a host, rename it the same way.

Most hosts are **TNFS** servers, and a plain name means TNFS. FujiNet can also reach other kinds of server if you spell the kind in front:

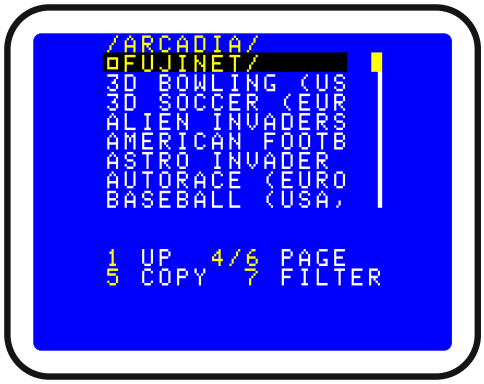
TYPE THIS	FOR
SD	the memory card in the cartridge
games.local or **192.168.1.20**	a TNFS server – the usual kind
smb://server/share	a Windows or Samba file share
nfs://server/export	an NFS share
ftp://server	an FTP server
http://server/path	a web server (**https://** too)

THE BROWSER

When you open a host, the browser shows what is in it, eight names to a page. The top line is the end of the folder you are in. Folders are in yellow and end in **/** they come first. The column of blocks at the right shows how far down the folder you are.



THE TOP OF APPS.IRATA.ONLINE



ITS ARCADIA FOLDER

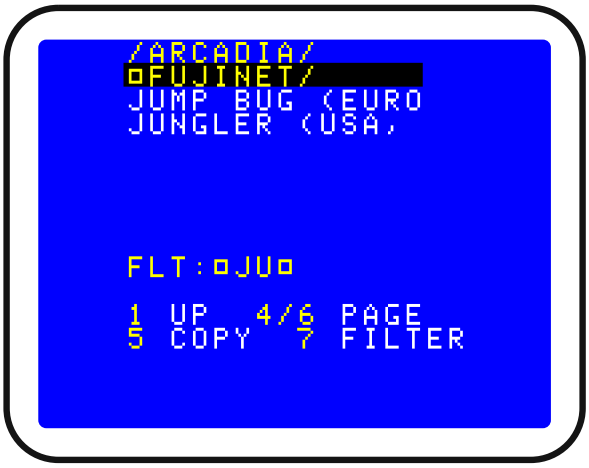
FIG. 18
THE BROWSER

IN THE BROWSER

▲▼ or 8	Move down and up the page. Past the bottom, go on to the next page.
◀▶ or 4 6	Turn the page back or on.
ACTION or ENTER	On a folder, open it. On a game, **load it** (Section 8).
1	Back up one folder.
5	Copy the game the bar is on to another host (Section 9).
7	Filter: show only the names that match a pattern.
CLEAR	Back to FujiNet hosts.

Names are cut to 14 characters; the full name is still what FujiNet loads. A character the ARCADIA-2001 cannot draw shows as a box: the **!** in front of **FUJINET/**, for one. An empty folder says **EMPTY FOLDER**.

Filters. Press 7 and spell a pattern on the character grid: ***** stands for any letters and **?** for any one letter, and the pattern must match the whole name. So ***JU*** shows only the names with JU in them, and **B*** those beginning with B. Capitals and small letters match each other, and folders are always shown. The pattern stays on the screen, after **FLT:**, until you leave the folder; an empty pattern shows everything again.

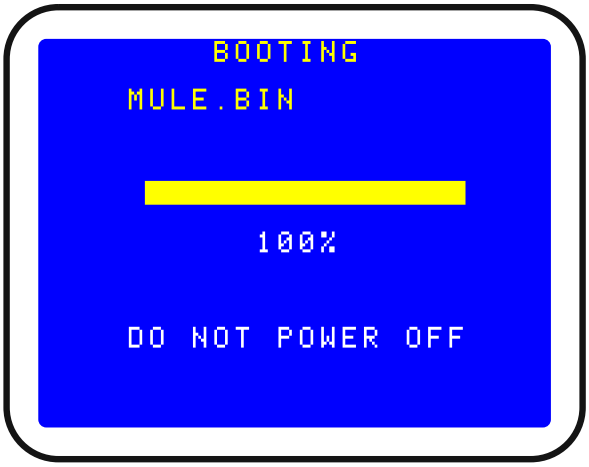


THE ARCADIA FOLDER, FILTERED BY **JU**

FIG. 19

8. LOADING GAMES

To load a game, find it in the browser and press the Action button (or ENTER). FujiNet sends the whole game into the cartridge and starts it – there is nothing else to do.



BOOTING

FIG. 20

BOOTING shows the game's name and a bar that fills as it loads. The game chip plays a short fanfare, the screen changes, and the game starts just as if its own cartridge were in the slot. **DO NOT POWER OFF** means just that, for the second or two it takes.

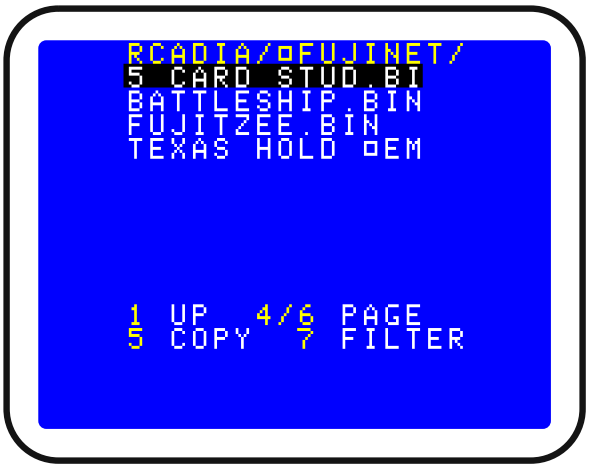
IMPORTANT: To leave a game and go back to CONFIG, press RESET on the back of the cartridge and then the console's RESET (Section 5). The console's RESET alone only starts the game over.

FROM THE INTERNET

The FujiNet project keeps the ARCADIA-2001's games on the Internet, ready to load. Nothing needs to be copied:

HOST	FOLDER	WHAT IS THERE
APPS.IRATA.ONLINE	**ARCADIA**	The ARCADIA-2001's own games: all 48 of them (Section 17)
APPS.IRATA.ONLINE	**ARCADIA ▸ !FUJINET**	The FujiNet network games: 5 Card Stud, Battleship, Fujitzee and Texas Hold'em (Sections 12–15)
APPS.IRATA.ONLINE	**MULE ▸ ARCADIA**	M.U.L.E. (Section 16)
TNFS.FUJINET.ONLINE		The FujiNet project's library for every computer and console

Open the host, open the folder, and choose the game.



ARCADIA ▸ !FUJINET: THE NETWORK GAMES

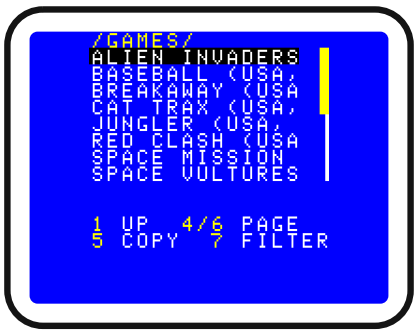
FIG. 21

FROM THE MEMORY CARD

1. On your computer, copy your games onto a microSD card formatted FAT32. You may put them in folders.
2. With the console OFF, push the card into the side of the cartridge until it clicks (Section 2).
3. Turn on. Open the **SD** host, and choose a game.



THE MEMORY CARD



A FOLDER ON IT

FIG. 22
GAMES ON THE MEMORY CARD

FROM YOUR OWN FILE SERVER

A computer in your home can serve games to FujiNet with a small free program called a **TNFS server**. There are versions for Windows, Mac, Linux and the Raspberry Pi; see **fujinet.online** for the current one.

1. Install the TNFS server, and point it at a folder of games.
2. Find your computer's network address – something like **192.168.1.20**.
3. On FujiNet hosts, rename an empty host to that address (Section 7).
4. Open it. Your folder appears in the browser.

A Windows share (**smb://**), an NFS share, or an FTP or web server works the same way.

WHICH GAMES WILL LOAD

FujiNet loads every game made for the ARCADIA-2001, up to 8,192 bytes – the size of the largest cartridge – and the larger programs made for FujiNet itself, up to 71,680 bytes. Of the 48 games in the library, 47 play; the one that does not is **Golf**, whose cartridge was wired differently from all the others. FujiNet does not imitate that cartridge yet.

File names. FujiNet loads files whose names end in **.BIN** or **.ROM**. A game file ending in **.ARC** loads once it is renamed to end in **.BIN**.

NOTE: FujiNet is made for the NTSC console. The games marked *Europe* in the library were sold for look-alike consoles in Europe, and play on the ARCADIA-2001 as well.

WHEN A GAME WILL NOT LOAD

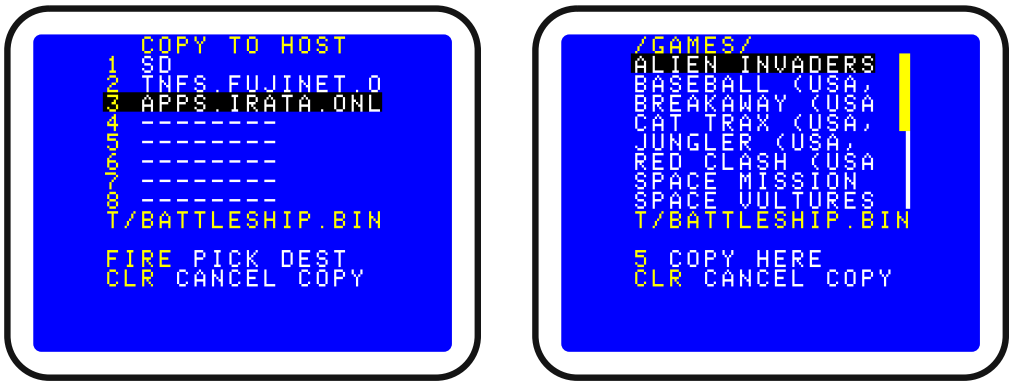
SCREEN SAYS	WHAT IT MEANS
NOT A CART	The file is not a game – or its name does not end in .BIN or .ROM.
ROM TOO BIG	Larger than 71,680 bytes.
STAGE ERROR	Larger than 8,192 bytes, but not a program made for FujiNet: the cartridge cannot place it.
TRUNCATED	The file was cut short on its way. Try again, or copy it again.
PATH ERROR, **MOUNT ERROR**	FujiNet could not open the host or the file. Check the host is running, and try again.

The message stays a moment, with a number at the right of the line for the FujiNet project should you ask them about it, and then you are back in the browser.

9. COPYING GAMES

FujiNet can copy a game from one host to another – from the Internet to your memory card, say, so that you can play it without a network.

1. In the browser, move the bar to the game you want to copy and press [5].
2. The list of hosts comes back, titled **COPY TO HOST**, with the game's name under it. Move to the host to copy to – **SD** for the memory card – and press the Action button.
3. Open the folder the game should go in, as in the browser.
4. Press [5] to copy it there.



STEP 2 STEP 3

FIG. 23
COPYING A GAME TO THE MEMORY CARD

The message line says **COPYING** while it works, and **COPIED** when it is done. Then you are back where you started, in the folder the game came from. To change your mind, press [CLEAR]: it cancels the copy.

If it says **COPY FAILED**, the memory card may be full or missing, or the host may not let you write to it – the Internet libraries, for one, are read-only. Folders cannot be copied (**FILES ONLY**), only games, one at a time. A small settings file kept beside a game, with the same name ending in **.CFG**, is copied with it; CONFIG does not list those files.

10. INFO AND THE WEB CONTROL PANEL

Press 3 on FujiNet hosts to see how FujiNet is connected:



LINE	WHAT IT IS
SSID	the WiFi network FujiNet joined
HOSTNAME	FujiNet's own name on your network
IP ADDRESS	FujiNet's address on your network
GATEWAY	your router's address
VERSION	the FujiNet firmware version

FIG. 24
FUJINET INFO

Press any button to go back.

THE WEB CONTROL PANEL

Everything FujiNet knows can also be seen and changed from a computer, tablet or phone on the same network. Type FujiNet's **IP ADDRESS** from the INFO screen into a web browser – for example **http://192.168.1.73/**. On many networks **http://fujinet.local/** works too.

The control panel lets you change the WiFi network, fill in or change the eight hosts with a real keyboard, put games on the memory card, and see the cartridge's settings and version. It is the easiest way to type a long host name, or a password with symbols in it.

SETTING WIFI FROM THE MEMORY CARD

If you would rather not spell a password with the hand controller at all, FujiNet can read its settings from the memory card. Make a plain text file called **fnconfig.ini** in the top folder of the card, holding:

```
[WiFi]
enabled=1
SSID=YourNetworkName
passphrase=YourPassword
```

Put the card in the cartridge and turn on. FujiNet reads the file each time it starts. (Everything you change in CONFIG or the control panel is written back to the same file, so the card keeps your hosts as well.)

11. THE GAME LOBBY

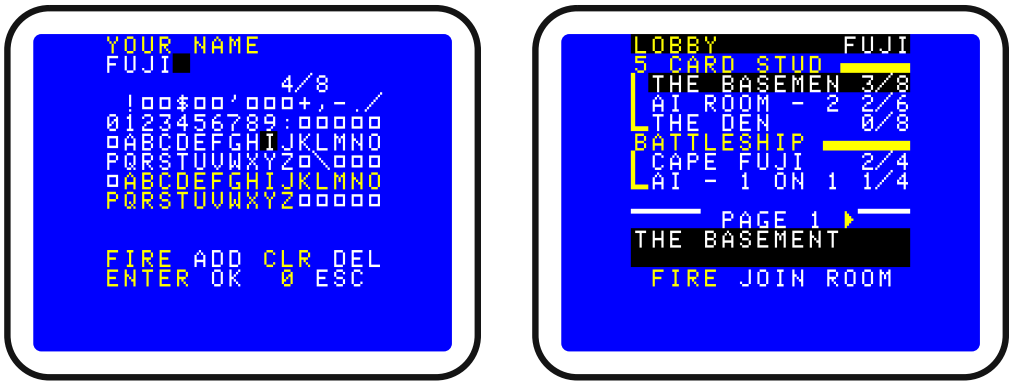
The **FujiNet Game Lobby** is a list of every network game that has a table open right now, on every kind of computer and console FujiNet works with. Pick a table and the Lobby loads the right game for your ARCADIA-2001 and sends you to it.

To open the Lobby, press [9] on FujiNet hosts. CONFIG says **GAME LOBBY**, looks for the Lobby's host, **ec.tnfs.io**, among your hosts – putting it in host 8 if it is not there – and loads the Lobby from it, as it loads any game.

NOTE: The ARCADIA-2001's Lobby is being finished now, and is not yet on ec.tnfs.io. Until it is, load the network games from the browser, as Section 8 shows: each has its own list of tables, and they work just the same without the Lobby.

YOUR NAME

The first time, the Lobby asks **YOUR NAME** on the character grid (Section 6): two to eight letters and numbers (**USE 2-8 A-Z 0-9**). Press [ENTER], and the Lobby remembers it.



YOUR NAME THE ROOMS

FIG. 25
THE GAME LOBBY

THE ROOMS

LOADING ROOMS, then the open tables, six to a page, under each game's name, with how many of their seats are taken. The bottom line takes turns telling you the buttons.

IN THE LOBBY

▲▼ or [8]	Move from table to table. Past either end, turn the page.
◀▶ or [4] [6]	Turn the page.
[ACTION] or [ENTER]	Join the table the bar is on: the Lobby loads the game (**JOINING**).
[0]	Fetch the list again, from the first page.
[CLEAR]	Change your name.

The **AI Rooms** seat you with robot players, the best way to learn a game; the others are for people, playing on an ARCADIA-2001, an Atari, an Apple, a PC or anything else FujiNet plugs into. If none are open the Lobby says **NO ROOMS OPEN** press [0] to look again.

12 5 CARD STUD

HOST APPS.IRATA.ONLINE **FILE ARCADIA/!FUJINET/5 Card Stud.bin**

I. GENERAL

Up to eight players around one table, on any computer or console FujiNet plays on – and if there are not enough people, the house's robots fill the seats.

1. Be sure the FujiNet cartridge is in the slot and FujiNet has joined your network (Section 6).
2. In the browser, open **APPS.IRATA.ONLINE**, then **ARCADIA**, then **!FUJINET**, and choose **5 CARD STUD.BIN**.
3. Play with the **left** hand controller.

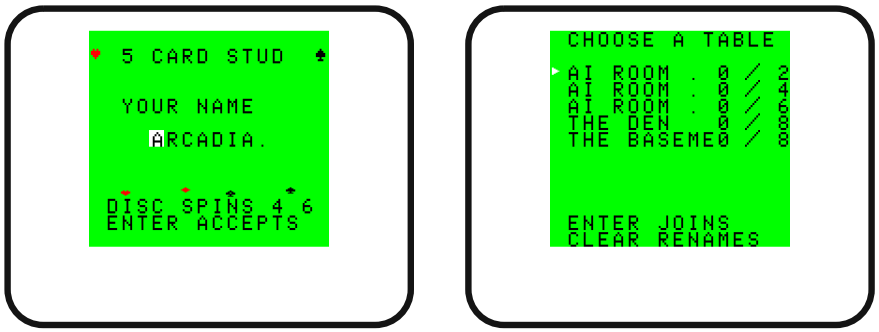
II. OBJECT OF THE GAME

Each hand you get one card face down and four face up, and the best poker hand takes the pot. Everyone starts with **200** chips.

1. Everyone puts **1** in the pot to start (the ante). Each player is dealt one card face down and one face up.
2. The player showing the lowest card must **POST** to begin the betting. Going round the table, each player may **CALL** (match the bet), **BET** or **RAISE** (make it bigger), **CHECK** (bet nothing, when there is nothing to call), or **FOLD** (drop out of this hand).
3. A third, fourth and fifth card are dealt face up, each followed by another round of betting. Bets are **5** in the early rounds and **10** later; a round allows no more than three raises.
4. When the betting is done, the hidden cards are turned over. The best poker hand wins the pot, and a few seconds later the next hand is dealt.

III. TO START THE GAME

1. The game asks **YOUR NAME**, with **ARCADIA** already spelled. Press the disc up or down to spin the letter at the cursor through A–Z, 0–9 and a blank; 4 and 6 move the cursor. Press ENTER.
2. **CHOOSE A TABLE**: each table with the players seated and the seats in all. **The Basement** and **The Den** are for people; the **AI Rooms** seat you with two, four or six robot players, which is the best way to learn.
3. Move to a table with the disc and press ENTER (**ENTER JOINS**). CLEAR goes back to your name (**CLEAR RENAMES**); 0 fetches the list again.



YOUR NAME THE TABLES

FIG. 26

IV. TO PLAY THE GAME

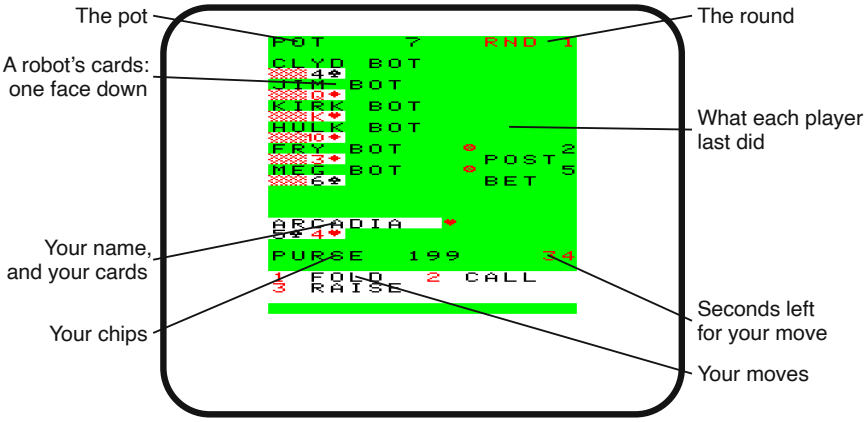


FIG. 27
YOUR TURN

Each seat shows the player's name over their cards. When it is your turn, your moves are numbered on the white line at the bottom – here **1 FOLD**, **2 CALL** and **3 RAISE** – with the seconds you have left at the right. Press the number of the move you want. Otherwise the line says **WAITING ON** and whose turn it is.

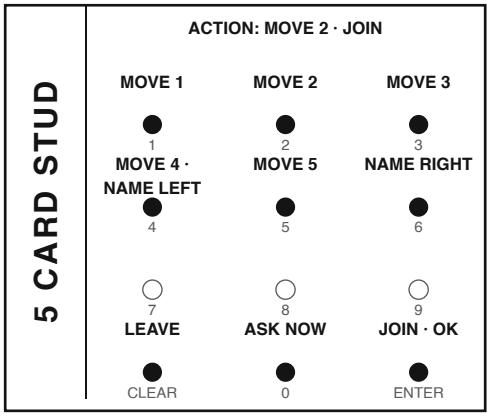


FIG. 28
OVERLAY

1. Watch the white line. When your moves appear, press [1] to [5] to make one. The Action button is move 2.
2. You have about **forty seconds** for each move. If the clock runs out, the server moves for you.
3. [0] asks the server for the table now, instead of waiting.
4. To leave the table, press [CLEAR]. Your seat is given back properly, so the others are not left waiting for you.

13 TEXAS HOLD'EM

HOST APPS.IRATA.ONLINE **FILE ARCADIA/!FUJINET/Texas Hold 'em.bin**

I. GENERAL

The other great poker game: two cards of your own, five shared by the whole table, and the best five-card hand wins. It plays like 5 Card Stud, on the same kind of table, with robots to fill the seats.

1. In the browser, open **APPS.IRATA.ONLINE**, then **ARCADIA**, then **!FUJINET**, and choose **TEXAS HOLD 'EM.BIN** (the apostrophe shows as a box).
2. Give your name and choose a table as in 5 Card Stud (Section 12). The tables are **The Basement**, **The Den** and the three **AI Rooms**.

II. OBJECT OF THE GAME

Everyone starts with **1,000** chips.

1. Two players put in the **blinds**, **5** and **10**, to start the pot. Everyone is dealt two cards face down.
2. Going round the table, each player may **CALL**, **RAISE** or **FOLD** – or **CHECK**, if there is nothing to call.
3. Three shared cards are dealt (the **flop**), then another betting round; then a fourth (the **turn**) and a fifth (the **river**), each with its own betting round.
4. At the **showdown**, each player makes the best five-card hand from their two cards and the five shared ones. The best hand takes the pot.

III. TO PLAY THE GAME

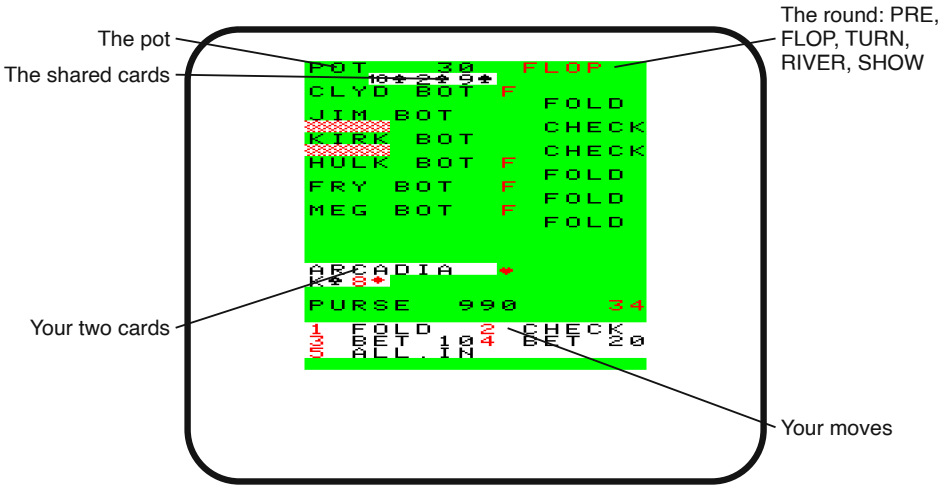


FIG. 29
AFTER THE FLOP

The shared cards lie in a row under the pot, with the round named at the right. Your two cards are at the bottom, under your name. Your moves – here **1 FOLD**, **2 CHECK**, **3 BET 10**, **4 BET 20** and 5, **ALL-IN** – are on the white line, with the seconds you have left. The buttons are 5 Card Stud's: 1 to 5 make a move (the Action button is move 2), 0 asks the server now, and CLEAR leaves the table.

14 BATTLESHIP

HOST APPS.IRATA.ONLINE **FILE ARCADIA/!FUJINET/Battleship.bin**

I. GENERAL

Two to four fleets on one ocean. Hide your five ships, then take turns firing – and every shot you fire lands on **every** enemy board at once.

1. In the browser, open **APPS.IRATA.ONLINE**, then **ARCADIA**, then **!FUJINET**, and choose **BATTLESHIP.BIN**.
2. Give your name and choose a table as in 5 Card Stud (Section 12). The **AI** tables give you one, two or three robot admirals; **Cape Fuji** and **High Seas** are for people.
3. The **LOBBY** lists the admirals at the table. Press the Action button when you are ready (**FIRE = READY**); press it again if you are not. The game starts when everyone is ready.

II. OBJECT OF THE GAME

Each fleet has five ships, five, four, three, three and two squares long, hidden on its own board. Sink every ship of every other fleet; the last fleet with a ship afloat wins.

III. TO START THE GAME

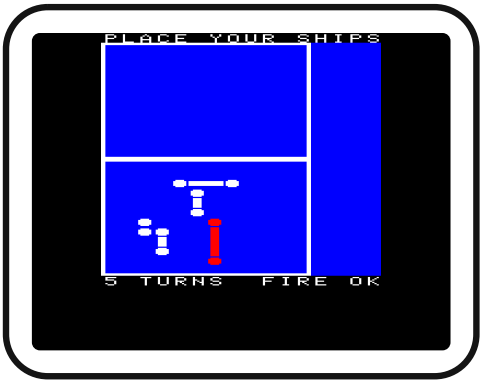


FIG. 30
PLACE YOUR SHIPS

1. **PLACE YOUR SHIPS**: the first ship appears on your board, at the bottom of the screen.
2. Move it with the disc. 5 turns it (**5 TURNS**).
3. Press the Action button to leave it there (**FIRE OK**). The next ship appears.
4. When all five are placed, the game says **WAITING ON SHIPS** until every admiral has placed theirs.

IV. TO PLAY THE GAME

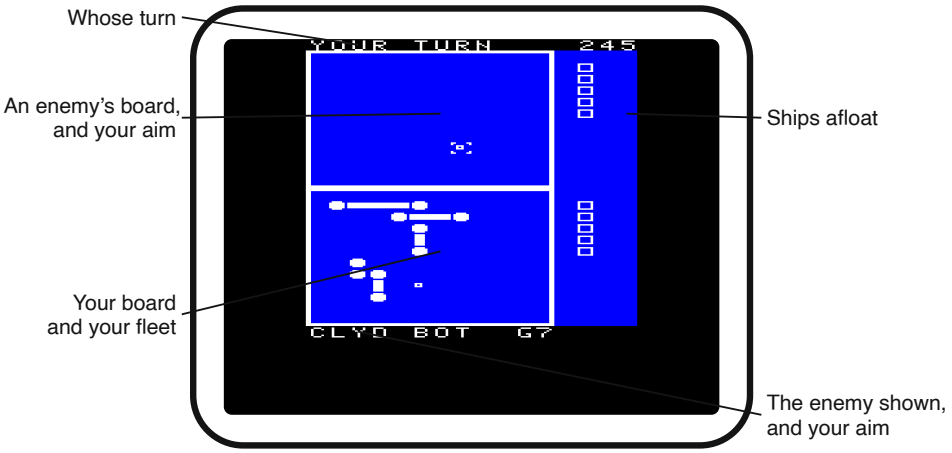


FIG. 31
YOUR TURN

The enemy's board is at the top, your own at the bottom, with each fleet's ships beside its board. When it is **YOUR TURN**, aim at any square and fire. The game tells you **MISS**, **HIT** or **SUNK**.

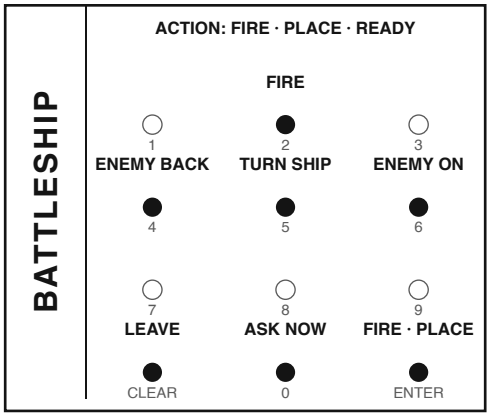


FIG. 32
OVERLAY

IN BATTLE

◀▲▼▶	Aim; while placing, move the ship.
ACTION or ENTER	Fire at that square, on every enemy board at once; place a ship.
4 6	Show the next or the last enemy's board, with three or four fleets.
5	Turn a ship, while placing.
0	Ask the server now, instead of waiting.
CLEAR	Leave the game.

15 FUJITZEE

HOST APPS.IRATA.ONLINE **FILE ARCADIA/!FUJINET/Fujitzee.bin**

I. GENERAL

The dice game. Roll five dice up to three times a turn, keep the ones you like, and score the result in one of the boxes on your card. Up to six players, with robots on hand.

1. In the browser, open **APPS.IRATA.ONLINE**, then **ARCADIA**, then **!FUJINET**, and choose **FUJITZEE.BIN**.
2. Give your name and choose a table as in 5 Card Stud (Section 12). **The Bar** and **Kitchen Table** are for people; the **AI Rooms** add two or four robots.
3. In the **LOBBY**, press the Action button when you are ready (**FIRE TO READY**); each player shows **READY** or **WATCH**. The game starts when everyone is ready.

II. OBJECT OF THE GAME

Fill every box on your card. After the last round the highest total wins.

BOX	SCORES
ONE through **SIX**	The total of the dice showing that number
UP, **BON**	Your top-half total, and a **bonus of 35** once it reaches 63
SET3, **SET4**	Three or four of a kind: the total of all five dice
HOUS	Three of one number and two of another: **25**
SRUN	Four in a row: **30**
LRUN	Five in a row: **40**
CNT	Anything: the total of all five dice
FUJI	All five the same: **50**
TOT	Your total

III. TO PLAY THE GAME

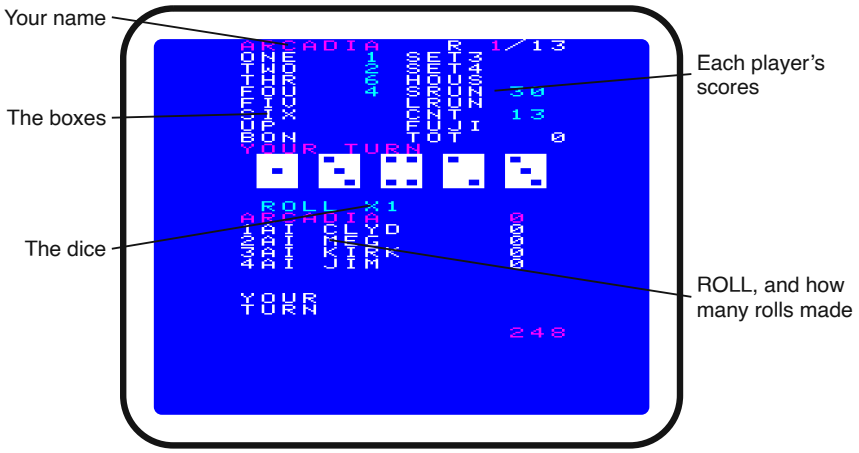


FIG. 33
YOUR TURN, AFTER THE FIRST ROLL

When it is **YOUR TURN**, the pointer is on **ROLL**.

1. Press the Action button to roll.
2. Move left and right along the dice. The Action button **keeps** a die; press again to let it go.
3. Move down to **ROLL** and roll again – up to three rolls in all.
4. Move up onto the card. Move to the box you want and press the Action button to score it. After the last roll the pointer goes to the box that pays most, so you may simply press the button.

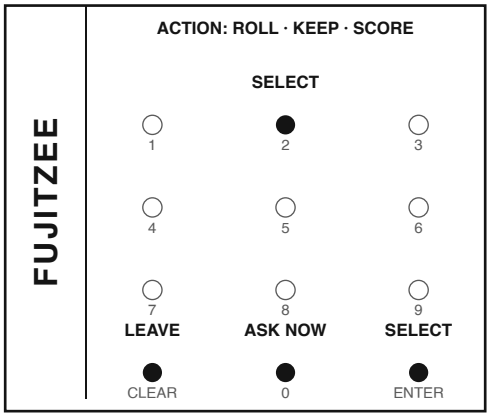


FIG. 34
OVERLAY

FUJITZEE

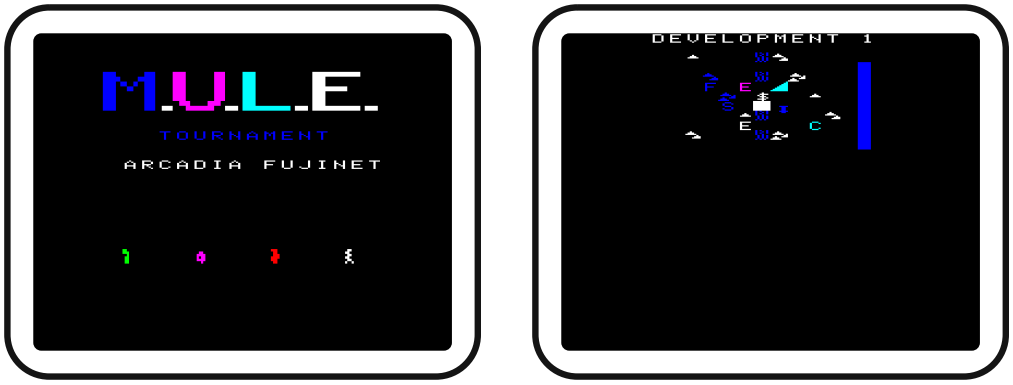
▲▼	Up and down the boxes; down to ROLL.
◀▶	Along the dice. While others play, turn to another player's card.
ACTION, 2 or ENTER	Ready; roll; keep a die; score a box.
0	Ask the server now.
CLEAR	Leave the table.

16 M.U.L.E.

HOST APPS.IRATA.ONLINE **FILE MULE/ARCADIA/mule.bin**

I. GENERAL

The classic game of settling the planet Irata, for four colonists on any FujiNet machine at all – the other three may be on an Atari, an Apple, a PC or another ARCADIA-2001, and the computer takes any empty seat. Twelve months to claim land, outfit your M.U.L.E.s, and trade your way to the richest colony.



THE TITLE THE MAP

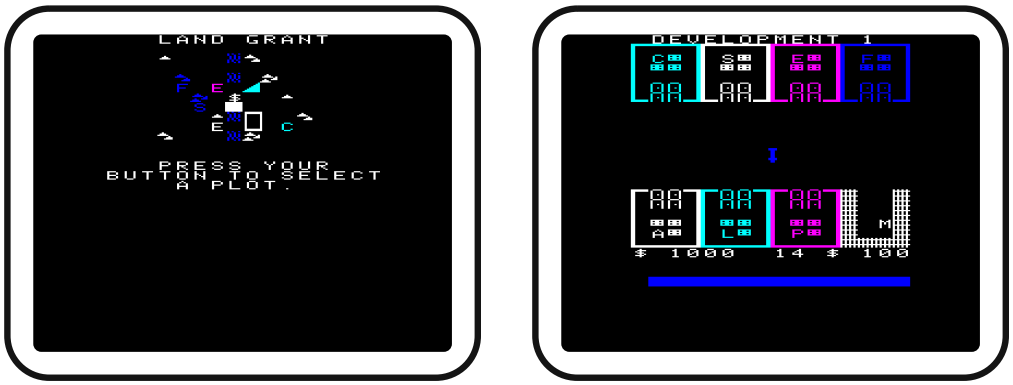
FIG. 35

1. In the browser, open **APPS.IRATA.ONLINE**, then **MULE**, then **ARCADIA**, and choose **MULE.BIN**. At 51,200 bytes it is the largest program the cartridge plays (Section 3).
2. Give your name: the disc up and down picks a letter, right takes it, left rubs it out; **ENTER** or the console's START when you are done.
3. Choose a table (**DISC, THEN FIRE**). In the colony room, choose your species with the disc and press the Action button when you are ready.

NOTE: If M.U.L.E. says it **NEEDS A FUJINET CART, PROTO 2.**, the cartridge's firmware is older than the game. Bring it up to date (Section 19).

II. TO PLAY THE GAME

M.U.L.E. is the one FujiNet game that uses the console's own buttons, too.



THE LAND GRANT THE TOWN

FIG. 36

M.U.L.E.

▲▼◀▶	Walk; choose; move your line in an auction; up to sell, down to buy.
ACTION or 2	Join and ready; claim a plot; begin your turn; install a M.U.L.E.; and every other "press the button" of the game.
START	End your turn. On the name screen, done.
OPTION	Leave the table (in a game it asks first: press OPTION again). The computer takes over your colonist.
SELECT	Sound off and on.
0	Sit out a declaration.
CLEAR	On the name screen, rub out a letter.

The whole game – the colonists, the land, every phase of a month, the auctions, every fortune and catastrophe – is in the *M.U.L.E. Player's Guide* from the FujiNet project.

17. THE CARTRIDGE LIBRARY

The ARCADIA-2001 was sold with a library of game cartridges “available at your EMERSON dealer”. The FujiNet project keeps all 48 of them on the Internet, ready to load: host **APPS.IRATA.ONLINE**, folder **ARCADIA** (Section 8). Each loads in a second or two, and plays just as its own cartridge would.

3D Bowling	Crazy Climber*	R2D Tank*
3D Soccer*	Crazy Gobbler*	Red Clash
Alien Invaders	Escape	Robot Killer*
American Football	Funky Fish*	Route 16*
Astro Invader*	Golf*†	Soccer
Autorace*	Grand Slam Tennis	Space Attack
Baseball	Hobo*	Space Mission
Basketball*	Horse Racing*	Space Raiders
Blackjack and Poker*	Jump Bug*	Space Squadron*
Boxing*	Jungler	Space Vultures
Brain Quiz	Missile War	Spiders
Breakaway	Monaco Grand Prix*	Star Chess
Capture	Nibblemen*	Super Gobbler*
Cat Trax	Ocean Battle	Tanks a Lot
Circus*	Parashooter*	The End*
Combat*	Pleiades*	Turtles*

* Sold in Europe, for look-alike consoles; they play on the ARCADIA-2001 too.
† Golf’s cartridge was wired unlike any other, and FujiNet does not imitate it yet: Golf stops at its first screen.

INSTRUCTIONS

Each game’s own instruction booklet tells how to play it – which overlay goes on which controller, and what START, SELECT and OPTION do. Most games use the console’s buttons the same way: SELECT picks the game number, OPTION the skill, and START starts or restarts a game.

To play another game, go back to CONFIG (Section 5) and choose it.

18. NO CONSOLE? FUJINET GO

FujiNet Go Arcadia 2001 Desktop is an ARCADIA-2001 for your computer – Windows, Mac or Linux – with a FujiNet cartridge already in its slot. It runs the same CONFIG, the same Lobby and the same games as the cartridge, from the same hosts, and needs no Arcadia of your own. It is free, from **github.com/FujiNetWIFI/fujinet-go-arcadia-desktop**, under Releases.

The computer's keyboard is the left hand controller:

ARCADIA	COMPUTER
Keyboard 1 to 9, 0	1 to 9, 0
CLEAR, ENTER	Backspace, Enter
Control disc	the arrow keys (or a game pad's stick)
Action button	Right Ctrl
START, OPTION, SELECT	F2, F3, F4
The console's RESET	F1
Back to CONFIG (the power switch)	Escape
The Controllers window	F9: both hand controllers, drawn; click a button, drag the disc

Games from the library load through CONFIG as on the console. A game kept on your computer opens with **File ▸ Open cartridge** and plays on its own cartridge – Golf's too.

Your own FujiNet. In Preferences, **External FujiNet** points the program at a FujiNet running elsewhere on your network, by its address and port, instead of the one built in.

19. MAINTAINING YOUR FUJINET CARTRIDGE

Your FujiNet cartridge should bring many years of enjoyment. To keep it in good working condition, please remember the following points:

– When the micro-computer is not in use, press its POWER switch to OFF.
– Always turn the POWER OFF before putting the cartridge in or taking it out.
– Do not try to force the cartridge into the console.
– Do not push anything into the three small service holes on the back of the cartridge.
– Push the memory card straight in and let it spring straight out. Do not pry at it.
– Use only a USB-C phone charger or a computer's USB port in the USB-C socket.
– Keep the cartridge away from heat and moisture.
– Do not clean the cartridge with detergents or solvents. For cleaning, use a slightly water-moistened cloth only.

KEEPING FUJINET UP TO DATE

FujiNet's firmware – the programs inside the cartridge – is improved all the time. New versions, and how to load them, are announced at **fujinet.online**. Loading new firmware does not change your hosts or your WiFi settings.

20. TROUBLE SHOOTING CHECK LIST

The following list of symptoms, causes and remedies is supplied to help you find the trouble when your FujiNet does not do what you expect. As a rule of thumb, if something does not work, turn the console OFF, wait a few seconds, and turn it ON again.

Symptom	Cause And Remedy
No playfield image on TV, or a whitish grey screen.	– Switch box set at “TV”. Change to “GAME”. – TV not on channel 3 or 4, or not the channel the console’s switch is set to. – Cartridge not firmly seated. Turn the console OFF, push the cartridge in firmly, and turn it ON. – Cartridge put in while the console was ON. Turn the POWER OFF and back ON.
NO FUJINET CART on the screen.	– The cartridge is not answering. Turn the console OFF, seat the cartridge firmly, and turn it ON. If it persists, press RESET on the back of the cartridge, then the console’s RESET.
The picture jumps when WiFi connects, or CONFIG starts over by itself.	– The console cannot give the cartridge enough power. Plug a USB-C phone charger into the top of the cartridge, before turning the console on (Section 4).
The hand controller does nothing.	– CONFIG listens to the left hand controller. Use that one.
My network is not in the SELECT NETWORK list.	– Router too far away. Move the console nearer, or the router. – The network uses only 5 GHz. FujiNet needs 2.4 GHz. – A hidden network. Choose OTHER NETWORK and spell its name. – Press [CLEAR] to listen again.
CONNECT FAILED.	– Wrong password. Small letters show in yellow; check each one (Section 6). – Or set your WiFi from a computer, or from the memory card (Section 10).
The status light is not white.	– FujiNet is not on your WiFi. See the two above.
A host shows nothing, or **HOST ERROR** or **FOLDER ERROR**.	– Host name misspelled. Rename it (Section 7). – The server is not running, or your Internet connection is down. – SD: no memory card, or it is not formatted FAT32.
NOT A CART.	– The file’s name does not end in .BIN or .ROM. Rename an .ARC file to .BIN (Section 8). – The file is not an ARCADIA-2001 game.
ROM TOO BIG, **STAGE ERROR** or **TRUNCATED**.	– A file too large, of a size the cartridge cannot place, or cut short. See Section 8.

Symptom	Cause And Remedy
I cannot get back to CONFIG.	– Press RESET on the back of the cartridge, then the console's RESET (Section 5). The console's RESET alone only starts the game over.
Golf stops at its first screen.	– Golf's cartridge was wired unlike any other; FujiNet cannot play it yet (Section 17).

NOTE: Due to differences in color circuitry on various makes of TV receivers, not all colors will appear the same on every TV set.

GETTING HELP

FujiNet is made by a worldwide community of volunteers, and they are glad to help:

– **fujinet.online** – news, downloads and the FujiNet wiki
– **github.com/FujiNetWIFI** – the source for everything, and the place to report a problem
– **discord.gg/7MfFTvD** – the FujiNet Discord, where you can ask anything

21. PARTS LIST

The FujiNet cartridge is open hardware: the board and every program in it may be built, changed and shared. Its design files are in **fujinet-hardware** on GitHub, under **Emerson-Arcadia / FujiNet-Arcadia-Rev0**. When asking about parts, give the board's name and revision, printed on the board.

Board	FujiNet-Arcadia Rev0
Game chip	RP2354A (RP2350 with 2 MB flash)
Network chip	ESP32-S3-WROOM-1, 16 MB flash, 8 MB PSRAM
WiFi	802.11 b/g/n, 2.4 GHz
Memory card	microSD, push-push, FAT32
USB	USB-C: power, and firmware loading
Lights	status (white / orange); game chip (red)
Buttons	RESET; three service buttons (BOOTSEL, S3 RST, S3 BOOT)
Power	from the cartridge slot, or USB-C (USB-C first)
Console	EMERSON ARCADIA-2001 (ADV-1), NTSC
Contacts	2 × 15, 2.54 mm
Games	up to 8,192 bytes; FujiNet programs up to 71,680; .BIN or .ROM
Cartridge shell	to come; the drawings in this manual are its own

RELATED PUBLICATIONS

– *ARCADIA-2001 Owner's Manual* (Emerson, 1982) – the console itself.
– *M.U.L.E. Player's Guide* – the whole of M.U.L.E.
– *The FujiNet Network Protocol Handbook* – everything FujiNet can talk to.

A FEW WORDS

WORD	MEANING
AI Room	A game table where robot players fill the empty seats.
CONFIG	The FujiNet program built into the cartridge: the hosts, the browser, WiFi and INFO.
Firmware	The programs built into the cartridge itself.
FAT32	The way most memory cards are formatted; the one FujiNet reads.
Host	A place games are kept: the memory card, or a server on the Internet or in your home.
Lobby	FujiNet's list of open game tables, on every kind of machine.
NTSC, **PAL**	The North American and the European ways of making a TV picture. The ARCADIA-2001 and FujiNet are NTSC.
Page	A 2K part of a large FujiNet program, shown to the console one at a time.
SSID	Your WiFi network's name.
TNFS	The simple file-sharing language FujiNet servers speak.
USB-C	The small, either-way-up plug of modern phone chargers.

MY FUJINET

Write down your settings here. If you ever start over, everything you need is on this page.

WIFI NETWORK (SSID) _____

PASSWORD _____

FUJINET'S IP ADDRESS _____

FIRMWARE VERSION _____

PLAYER NAME _____

HOSTS

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

GAMES I LIKE, AND WHERE THEY ARE

_____

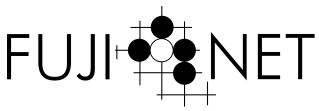
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QUICK REFERENCE

SCREEN	DISC	ACTION, 2, ENTER	KEYBOARD	CLEAR
THE FUJINET CONFIG (LEFT HAND CONTROLLER)				
Select network	choose	this one	0 skip WiFi	listen again
Character grid	move (or 4 6 8)	type · ENTER: OK	0 leave	erase
FujiNet hosts	choose (or 8)	open	1 rename · 3 info · 7 WiFi · 9 Lobby	–
Browser	choose; left, right: page	open · load	1 up · 4 6 page · 5 copy · 7 filter	to hosts
Copy to host	choose	pick · open	5 copy here	cancel
Game Lobby	choose; left, right: page	join	0 list again	your name
THE GAMES				
5 Card Stud, Texas Hold'em	choose a table	join · move 2	1 to 5 your move · 0 ask now	leave
Battleship	aim; move a ship	fire · place · ready	5 turn · 4 6 enemy · 0 ask now	leave
Fujitzee	boxes; dice; ROLL	roll · keep · score	0 ask now	leave
M.U.L.E.	walk · choose · bid	the button	0 sit out · START end turn · OPTION leave · SELECT sound	rub out
EVERYWHERE				
Back to CONFIG	RESET on the back of the cartridge, then the console's RESET.			
Console RESET	Starts the game you are playing over.			



A FujiNet Project Publication
fujinet.online · github.com/FujiNetWIFI

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